



Scoring Guidelines

Scoring Guidelines for Question 1:
Mathematical Routines

10 points

Learning Objectives: 1.3.A 2.5.A 2.6.A 2.7.A 5.2.A 5.3.B 5.6.A

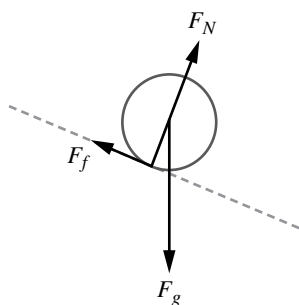
(a) i For a force of gravity drawn straight down from the center of the circle 1 point

Scoring Note: Acceptable labels for forces include: F_g , F_{gravity} , mg , $F_{\text{earth,disk}}$, W G , and g are not acceptable labels for the force of gravity.

For one of the following: 1 point

- F_N points perpendicular to the ramp starting from the point on the circle that touches the ramp.
- F_f points up the ramp starting at the point on the circle that touches the ramp.

Example Response:



(a) ii For indicating that the torque exerted on the disk around the center of the disk is from the frictional force 1 point

(a) iii For an expression for net force that includes a component of the gravitational force and frictional force in opposite directions. $mg \sin \theta - F_f$ 1 point

(a) iv For a multi-step derivation that uses Newton's second law in rotational or translational form. 1 point

For using the relationship between the angular acceleration of the disk and the translational acceleration of the center of mass of the disk. $\alpha = \frac{a}{R}$ 1 point

For a correct expression for the translational acceleration of the center of mass of the disk in terms of given variables 1 point

$$a = \frac{2}{3}g(\sin \theta)$$

Total for part (a) 7 points

Example Response:

$$\sum \tau = I\alpha$$

$$F_f R \sin(90^\circ) = \left(\frac{1}{2} MR^2\right) \alpha$$

$$F_f R \sin(90^\circ) = \left(\frac{1}{2} MR^2\right) \left(\frac{\alpha}{R}\right)$$

$$F_f = \frac{1}{2} Ma$$

$$\sum F = Ma$$

$$Mg \sin \theta - F_f = Ma$$

$$Mg \sin \theta - \frac{1}{2} Ma = Ma$$

$$Mg \sin \theta = \frac{3}{2} Ma$$

$$a = \frac{2}{3} g \sin \theta$$

(b) For a claim that $t_{\text{disk}} > t_{\text{block}}$ with an attempt at a justification using forces or conservation of energy. **1 point**

For indicating that the acceleration or the net force exerted on the disk is smaller than the acceleration or the net force exerted on the block. **1 point**

OR

For indicating that the disk will have both rotational and translational kinetic energy at the bottom of the ramp

For a justification that includes that the larger acceleration for the block over the same distance leads to a smaller time for the block to reach the bottom of the incline **1 point**

OR

For a justification that includes that the larger translational kinetic energy for the block over the same distance leads to a higher velocity and thus a shorter time over the same distance.

Example Response

The net force exerted on the block is larger since there is no frictional force from the surface. The net force is related to the acceleration, so since the block has a larger net force it will have a larger acceleration. Both the block and the disk start from rest and have to travel the same distance, and since the block has a larger acceleration, it will be able to travel the distance in a smaller time. Therefore, the block will reach the bottom in less time.

Total for part (b) 3 points

Total for question 1 10 points

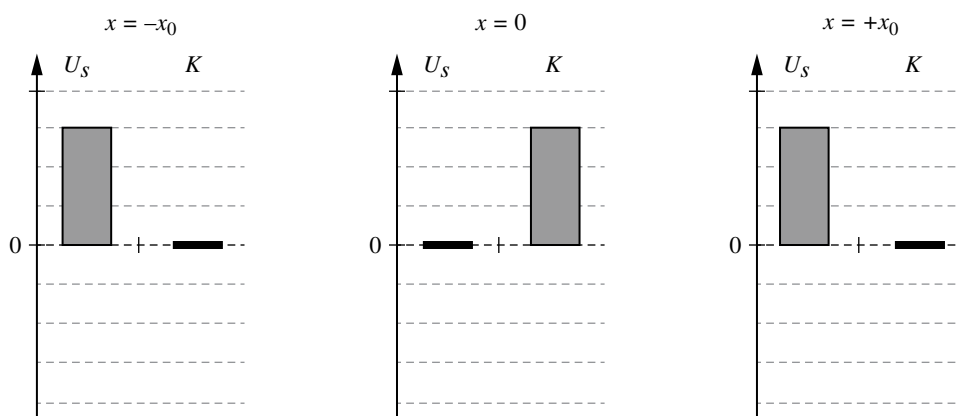
Scoring Guidelines for Question 2: Translation Between Representations

12 points

Learning Objectives: 1.3.A 2.8.A 3.4.B 7.2.A 7.3.A 7.4.A

- | | | |
|-----|---|---------|
| (a) | For a single bar representing the kinetic energy of the block at $x = 0$. | 1 point |
| | For a single bar representing the potential energy of the block-spring system at $x = +x_0$. | 1 point |
| | For an equal total energy of the block-spring system at all three locations. | 1 point |

Example Response



Total for part (a) 3 points

- | | | |
|-------|--|---------|
| (b) i | For the correct spring constant in terms of $F_{\max}$ and x_0 .
$k = \frac{F_{\max}}{x_0}$ | 1 point |
|-------|--|---------|

- | | | |
|--------|---|---------|
| (b) ii | For a multi-step derivation that begins with the equation for period of a mass on a spring.
$T = 2\pi\sqrt{\frac{m}{k}}$ | 1 point |
|--------|---|---------|

- | | |
|---|---------|
| For ONE of the following: | 1 point |
| <ul style="list-style-type: none"> • Substitution of the spring constant into the equation for the period of a block-spring system. • Substitution of $2t_0$ for the period into the equation for the period of a block-spring system. | |

- | | |
|---|---------|
| For a correct expression of the mass of the block in terms of given variables.
$m = \left(\frac{t_0}{\pi}\right)^2 \frac{F_{\max}}{x_0}$ | 1 point |
|---|---------|

Example Response:

$$k = \frac{F_{\max}}{x_0}$$

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$2t_0 = 2\pi\sqrt{\frac{m}{k}}$$

$$2t_0 = 2\pi\sqrt{\frac{m}{\frac{F_{\max}}{x_0}}}$$

$$t_0 = \pi\sqrt{\frac{x_0 m}{F_{\max}}}$$

$$\left(\frac{t_0}{\pi}\right)^2 = \frac{x_0 m}{F_{\max}}$$

$$m = \left(\frac{t_0}{\pi}\right)^2 \frac{F_{\max}}{x_0}$$

Total for part (b)**4 points**

- (c) For a sketch that has the same period as the graph given in part (b) of $2t_0$.

1 point

For a sketch that has the same magnitude above and below the zero.

1 point

For a sketch that resembles a positive sine function.

1 point**Example Response**

Velocity (m/s)

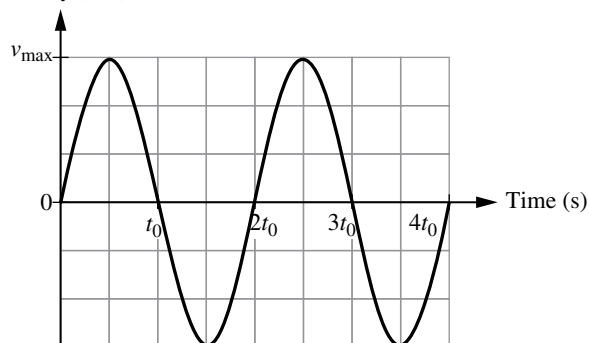


Figure 4

Total for part (c)**3 points**

- (d) For a claim that the sketch and claim are or are not consistent with the graph of velocity as a function of time, consistent with part (c) with an attempt at a justification using functional dependence.

1 point

For a justification that includes that at $t = 1.5t_0$, the net force should be zero with justification using a relevant feature of the graph (i.e., the slope of the graph of velocity as a function of time is zero at $t = 1.5t_0$).

1 point

Example Response:

The claim and free-body diagram are not consistent with my graph of velocity as a function of time. At time $t = 1.5t_0$, my velocity graph is at a maximum, and the slope of the velocity graph at that time is zero. This means that at $t = 1.5t_0$, the acceleration is zero, and because acceleration is directly related to force, the force from the spring should be zero at $t = 1.5t_0$.

Scoring Note: The two points for Part (d) can be earned independently of the rest of the points on the question, as long as the student correctly interprets the consistency between their answer for part (c) and the given sketch. For example, a student who earns zero points in part (c) can still earn full credit in (d) if they reason why their incorrect diagram is or is not consistent with the sketch.

Total for part (d)	2 points
--------------------	----------

Total for question 2	12 points
----------------------	-----------

Scoring Guidelines for Question 3: Experimental Design and Analysis

10 points

Learning Objectives: **1.5.B** **8.2.B** **8.4.B**

(a) i For indicating all the required quantities to determine the density of the fluid – Pressure and height of the fluid. **1 point**

(a) ii For a reasonable method of reducing experimental uncertainty either by taking several pressure measurements at the same depth/height, or at least one pressure measurement at several depths/heights. **1 point**

Example Response

Place the pressure sensor at the bottom of the container of liquid. Measure the height of the liquid with a meterstick and record the reading on the pressure sensor. Repeat measurements to reduce experimental error. Put more liquid in the container and measure the new height of the liquid and the pressure from the sensor. Repeat for multiple heights of the liquid to reduce experimental uncertainty.

Total for part (a) **2 points**

(b) i For describing a graph that would have a slope that is either directly or inversely proportional to the density of the fluid. Examples of acceptable responses may include the following: **1 point**

- P as a function of h
- h as a function of P

(b) ii For describing how the slope of the graph can be used to determine the density. Examples of acceptable responses may include the following: **1 point**

- if Pressure as a function of height is graphed, slope is ρg
- if Height as a function of pressure is graphed, slope is $\frac{1}{\rho g}$
- if $\frac{P}{\rho}$ as a function of height is graphed, the slope is g
- if Height as a function of $\frac{P}{\rho}$ is graphed, the slope is $\frac{1}{g}$
- if P as a function of ρh is graphed, the slope is g
- if ρh as a function of P is graphed, the slope is $\frac{1}{g}$

Example Response:

Graph Pressure as a function of height, the slope of the best fit line of the data will be equal to ρg .

Total for part (b) **2 points**

(c) i For labeling the vertical axis with a quantity (including units) that could be plotted to produce a linear graph whose slope can be used to determine the acceleration due to gravity. Examples of acceptable responses may include the following: **1 point**

- Graphing v^2 as a function of Δy
- Graphing v^2 as a function of $2\Delta y$

Scoring Note: Any response that correctly identifies the functional dependence between speed and distance earns this point, regardless of any coefficients that contain numbers or physical/fundamental constants.

- (c) ii For labeling the vertical axis with a linear scale. **1 point**

Scoring Note: This point can be earned for labels and plotted points that are consistent with the previous part.

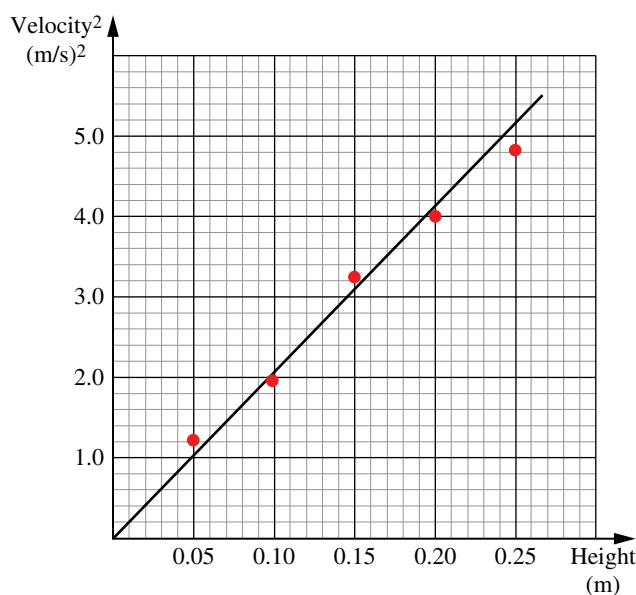
For correctly plotting data points consistent with quantities indicated in C (i). **1 point**

Scoring Note: This point can be earned for labels and plotted points that are consistent with the previous part.

- (c) iii For drawing an appropriate best-fit line that approximates the trend of the data. **1 point**

Scoring Note: If the graph produced is nonlinear, then a line or curve that approximates the trend of that data can earn this point.

Example Response:



Total for part (c) 4 points

- (d) For correctly relating the slope of the best-fit line to the acceleration due to gravity, g . Examples of acceptable responses may include the following: **1 point**

Graph	Analysis
v^2 vs. h	Slope equals $2g$
$\frac{1}{2}v^2$ vs. h	Slope equals g
v vs. $\sqrt{2h}$	Slope equals $\sqrt{g}$
v vs. $\sqrt{h}$	Slope equals $\sqrt{2g}$

For a value of g within a range 9 m/s^2 to 12 m/s^2 **1 point**

Example Response:

Slope equals $2g$

$$\text{slope} = \frac{1.0 - 0.0}{0.05 - 0.0} = 20 \frac{\text{m}}{\text{s}^2}$$

$$v^2 = 2gh$$

$$g = \frac{\text{slope}}{2} = \frac{20}{2} = 10 \frac{\text{m}}{\text{s}^2}$$

Total for part (d) 2 points

Total for question 3 10 points

Scoring Guidelines for Question 4: Qualitative/Quantitative Translation

8 points

Learning Objectives: **4.1.A** **4.3.A**

(a) For a claim that the final velocity will be approximately equal to the initial velocity of Block 1. **1 point**

$$v_f \approx v_0$$

For connecting the claim to **ONE** of the following: **1 point**

- the conservation of momentum.
- Newton's third law
- the momentum of the center of mass of the system

For the justification of the evidence supporting the claim. Examples of acceptable responses may include the following: **1 point**

- a correct statement relating either block's mass to the relative magnitude of the change in that block's momentum.
- a correct statement relating Newton's third law to a small acceleration for Block 2 or a large acceleration for Block 1
- a correct statement relating the speed of either block to the momentum of the center of mass of the system.

Example Response

If the mass of Block 2 is really really small, the final momentum of Block 2 will also be really, really small. This means that the change in momentum of Block 2 is really small. The change in momentum of each block is equal and opposite, so if the change in momentum of Block 2 is small, so is the change in momentum of Block 1, so Block 1 is still traveling at approximately the same speed.

OR

Newton's third law states that the force that the two blocks exert on each other must be equal in magnitude and opposite in direction. If the mass of Block 2 is very very small, the force exerted on it by Block 1 will mean that relative to Block 1, Block 2 will have a large acceleration (changing it's velocity from zero to almost v_0), while that same force exerted on Block 1 will result in a very small acceleration – so the speed of Block 1 remains almost equal to v_0 .

OR

If the mass of Block 2 is very very small, the addition of the mass of Block 2 to the mass of the system is a very small mass change. Which means that the additional momentum of adding the momentum of Block 2 moving is very very small, which means that the final speed of the system should be equal to the initial speed of the system because of the conservation of momentum.

Total for part (a) 3 points

(b) For a multistep derivation that begins with conservation of momentum **1 point**

$$p_i = p_f$$

For substituting appropriate values of mass and initial velocities **1 point**

$$M_1 v_{1,i} + M_2(0) = (M_1 + M_2) v_f$$

For a correct equation for the final velocity of the two-block system after the collision. **1 point**

$$v_f = \frac{M_1 v_0}{(M_1 + M_2)}$$

Example Response:

$$p_i = p_f$$

$$M_1 v_{1,i} + M_2 v_{2,i} = (M_1 + M_2) v_f$$

$$M_1 v_{1,i} + M_2 (0) = (M_1 + M_2) v_f$$

$$v_f = \frac{M_1 v_0}{(M_1 + M_2)}$$

Total for part (b) **3 points**

-
- (c) For a statement that addresses functional dependence between the mass of Block 2 and the final speed of the two-block system. **1 point**

Scoring Note: It is not necessary to use functional dependence correctly to earn this point.

For a correct justification using functional dependence to link the claim made in part (a) to the equation derived in part (b). **1 point**

Example Response:

In Part (a) I claimed that if the mass of Block 2 was much less than the mass of Block 1 that the final speed of the two-block system would be approximately equal to the original speed of Block 1. In the equation I derived in part (b) since the final speed of the block is inversely related to the mass of Block 2, if the mass of Block 2 is very small, it will effectively mean that the masses in the numerator and denominator cancel leaving the final mass equal to the initial mass, which is what I claimed in part (a).

Total for part (c) **2 points**

Total for question 4 **8 points**